

BILBAO AP, Spain

WMO: 080250

Lat: 43.298N

Lon: 2.906W

Elev: 138

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.7	34.0	24.7	18.9	41.6	27.9	22.0	41.0	27.5	53.3	24.4	53.6	4.0	90	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.5	89.9	69.5	85.5	68.5	81.9	67.5	72.9	83.6	71.1	80.4	69.7	77.7	8.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	110.5	76.2	68.2	104.1	74.5	66.8	99.0	73.1	36.6	84.2	34.9	80.7	33.7	77.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.9	18.5	15.9	DB	27.4	101.4	3.0	4.9	25.2	104.9	23.5	107.8	21.8	110.6	19.7	114.1
			WB	26.4	76.3	3.0	2.0	24.3	77.8	22.5	78.9	20.8	80.0	18.6	81.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	49.1	49.4	53.2	55.8	61.1	66.6	70.0	70.7	67.1	62.9	54.3	50.4	(6)
	DBStd	9.79	6.01	6.40	6.47	5.95	6.00	5.74	4.64	4.71	5.39	6.71	6.21	6.51	(7)
	HDD50	321	88	81	37	12	1	0	0	0	0	1	26	75	(8)
	HDD65	2728	492	438	370	284	153	45	7	4	36	122	324	453	(9)
	CDD50	3705	61	63	136	186	344	499	619	642	513	400	155	87	(10)
	CDD65	635	0	0	4	7	31	93	161	181	99	56	3	0	(11)
	CDH74	3947	0	3	35	86	276	636	904	1064	626	309	8	0	(12)
	CDH80	1240	0	0	3	23	92	246	276	344	196	60	0	0	(13)
(14) Wind	WSAvg	6.1	6.5	6.8	7.1	6.7	6.4	5.8	5.6	5.2	5.0	5.5	6.2	6.2	(14)
(15) Precipitation	PrecAvg	46.2	5.4	5.0	3.8	4.0	3.0	2.4	2.0	2.0	2.5	3.5	6.9	4.8	(15)
	PrecMax	65.0	11.9	13.6	8.5	8.7	7.0	9.4	4.6	6.1	7.4	9.0	13.5	9.6	(16)
	PrecMin	35.7	0.2	1.4	0.5	1.3	1.1	0.6	0.7	0.3	0.4	0.6	2.9	0.4	(17)
	PrecStd	7.2	3.1	3.6	2.3	2.4	1.4	2.1	1.1	1.5	1.5	2.2	2.5	2.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.2	71.7	78.7	82.7	89.4	96.7	95.2	95.3	91.8	85.9	74.7	69.7	(19)
		MCWB	55.5	54.6	58.0	61.2	66.9	71.0	71.7	71.7	68.8	65.5	60.8	54.9	(20)
	2%	DB	64.5	66.5	72.7	74.9	81.0	86.3	86.4	88.0	85.8	80.8	70.2	66.0	(21)
		MCWB	53.2	53.1	56.3	58.5	63.9	68.6	70.7	70.6	67.9	64.3	58.3	54.6	(22)
	5%	DB	61.6	62.9	68.2	70.0	75.4	80.5	82.1	82.8	80.6	77.1	67.5	62.7	(23)
		MCWB	52.1	51.6	54.4	57.3	62.1	67.1	68.9	69.5	66.9	62.9	57.0	52.9	(24)
	10%	DB	58.9	59.4	64.5	66.2	71.4	75.9	78.7	79.1	76.9	73.7	64.2	60.4	(25)
		MCWB	50.3	50.0	53.0	55.5	60.1	65.2	67.9	68.2	65.7	61.4	55.4	51.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	57.3	60.4	63.5	68.9	73.3	75.1	76.3	72.5	68.6	63.3	58.1	(27)
		MCDB	65.1	67.2	73.9	78.9	84.1	89.5	88.0	83.6	84.8	78.1	70.7	64.4	(28)
	2%	WB	55.1	54.8	57.4	60.2	65.5	70.3	72.4	73.4	70.2	66.4	60.0	56.1	(29)
		MCDB	61.7	62.8	68.9	71.6	77.1	81.7	82.3	82.9	79.9	75.5	66.7	62.4	(30)
	5%	WB	53.4	53.1	55.7	58.3	63.4	68.4	70.6	71.3	68.6	64.8	58.3	54.4	(31)
		MCDB	59.4	60.0	65.5	67.6	72.6	77.7	78.7	79.7	77.1	73.7	65.0	60.7	(32)
	10%	WB	51.8	51.5	54.1	56.5	61.5	66.6	69.1	69.7	67.1	63.2	56.6	52.8	(33)
		MCDB	57.1	57.6	62.5	64.0	69.1	74.2	76.1	76.9	74.4	71.1	62.5	58.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	15.0	16.7	16.9	16.9	16.6	16.4	17.5	18.1	17.2	13.6	13.7	(35)
		MCDBR	16.9	21.2	23.9	24.7	26.2	26.5	24.5	25.1	25.2	22.4	18.2	16.0	(36)
	5% WB	MCWBR	9.8	11.6	11.9	12.4	12.3	11.9	10.8	11.3	11.1	10.3	9.7	9.1	(37)
		MCWBR	14.8	18.2	21.1	21.8	22.8	22.4	20.5	21.7	21.6	19.7	15.7	14.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.329	0.343	0.376	0.398	0.410	0.423	0.413	0.411	0.390	0.366	0.344	0.325	(40)
		taud	2.505	2.452	2.363	2.313	2.314	2.314	2.345	2.361	2.412	2.480	2.495	2.514	(41)
	Ebn at Noon		255	270	273	275	274	270	272	268	266	258	246	245	(42)
		Edh at Noon	23	28	34	39	40	40	39	37	32	27	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		443	698	1089	1443	1723	1892	1911	1683	1348	876	515	415	(44)
	RadStd		53	95	136	148	126	127	92	100	68	74	63	49	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page